

**Department of Computer & Mathematical Sciences
University of Toronto Scarborough**

**MATA31: Calculus I for Mathematical Sciences
Final Examination**

Examiner: N. Breuss

Date: December 14, 2017

Duration: 3 hours

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO.

FAMILY NAME: _____

GIVEN NAMES: _____

STUDENT NUMBER: _____

SIGNATURE: _____

NOTES:

- NO calculators, or any electronic aid is permitted. If you have a cell phone in the room, make sure it is away from you and **turned off**.
- No books, notebooks or scrap paper are permitted near your examination table.
- There are 17 numbered page in the test. It is your responsibility to ensure that, at the start of the test, this booklet has all its pages. Please leave all the pages of this booklet stapled. Do not remove any pages.
- Answer all questions in the space provided. **Show your work and justify your answers for full credit.**

Question	Marks
1	/3
2	/5
3	/2
4	/8
5	/24
6	/4
7	/4
8	/4
9	/10
10	/5
11	/8
12	/7
13	/4
14	/12
Bonus 1	/5
Bonus 2	/8
Bonus 3	/12
TOTAL	/100

1. [3 points]

For what value of $\mathbf{a}$ and $\mathbf{b}$ is $f(x) = \begin{cases} ax + 2b, & x \leq 0 \\ x^2 + 3a - b, & 0 < x \leq 2 \\ 3x - 5, & x > 2 \end{cases}$ continuous at every $\mathbf{x}$?

2. [5 points]

Carefully state and prove the Addition Law for limits.

3. [2 points]

State, without proof, the Squeeze Theorem for limits.

4. [8 points]

Use Squeeze Theorem to prove $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

5. [24 points]

Use the Limit Laws to evaluate the following limits:

a) $\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 2x})$

b) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$

c) $\lim_{x \rightarrow 0} \frac{\arctan x}{\arcsin x}$

d) $\lim_{x \rightarrow \infty} \left(\frac{x}{x-1} \right)^x$

6. [4 points]

Show that $(x^n)' = nx^{n-1}$ if n is a positive integer.

7. [4 points]

Use implicit differentiation to find an equation of the tangent line to the curve $y \sin 2x = x \cos 2y$ at point $(p/2, p/4)$.

8. [4 points]

Use linear approximation to find $\ln(1.01)$

9. [10 points]

Find the derivatives of the following functions:

a) $f(x) = \sqrt{\frac{x^2 + 1}{\sqrt[3]{x} - 1}}$

b) $f(x) = \cos x^{\sin\left(\frac{1}{\sqrt{x}}\right)}$

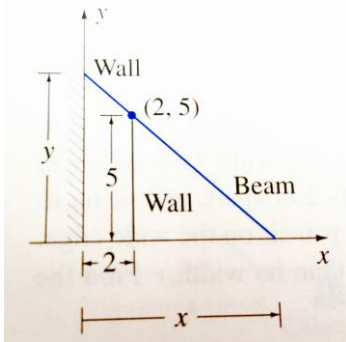
10. [5 points]

Find the derivative of $f(x)$ and/or prove that it doesn't exist.

c) Given $f(x) = \begin{cases} 3x+1, & x \leq 1 \\ x^3, & x > 1 \end{cases}$

11. [8 points]

A wall of a building is to be braced by a beam that must pass over a parallel wall 5m high and 2 m from the building. Find the length of the shortest beam that can be used if dimensions of the beam and the 5m wall are ignored.



12. [7 points]

Carefully state and prove Lagrange's Theorem (Mean Value Theorem).

13. [4 points]

Suppose that $f(x)$ is differentiable on $(1, 3)$, continuous on $[1, 3]$ and $f(1) = 2$.

Given that $1 \leq f'(x) \leq 5$ for all $x \in (1, 3)$.

What is the least value that f can take on at 3 ?

What is greatest value that f can take on at 3 ?

14. [12 points]

Sketch the graph of the function $f(x) = \frac{2+x-x^2}{(x-1)^2}$

a) [1 point] domain

b) [1 point] intercepts

c) [1 point] symmetry

d) [2 points] asymptotes (vertical, slant, horizontal)

e) [2 points] derivatives

f) [1 point] critical points and their classification

g) [1 point] inflection points

h) [2 points] number lines for f, f', f''

i) [1 point] final graph

BONUS

1. [5 points] Show $\cosh x = \ln(x + \sqrt{x^2 - 1})$ for $x \geq 1$

2. [8 points]. Find the antiderivatives of the following functions.

a) $f(x) = \cot^2 x$

b) $f(x) = \frac{1}{4 + 2x^2}$

3. [12 points] Sketch the graph of $f(x) = \arctan\left(\frac{x-1}{x+1}\right)$. Carefully indicate domain, points of X and Y intercepts, asymptotes, critical points, concavity.

